

Matrix Operations at a Bike-Share Depot

Answer Key

Algebra 2

Quick Answers

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|---|--|
| 1. row 1, column 1 = 43, row 1, column 2 = 40,
row 2, column 1 = 58, row 2, column 2 = 27 | row 2, column 1 = -6, row 2, column 2 = 11 |
| 2. row 1, column 1 = 7.5, row 1, column 2 =
12.5, row 2, column 1 = 6.25, row 2, column 2
= 11.25 | 6. row 1, column 1 = 16, row 1, column 2 = 15,
row 2, column 1 = 18, row 2, column 2 = 23 |
| 3. row 1, column 1 = 7, row 1, column 2 = 4,
row 2, column 1 = 6, row 2, column 2 = 3 | 7. $x = 5$, $y = 4$ |
| 4. North depot = 320, South depot = 332 | 8. (a) = 59, — |
| 5. row 1, column 1 = 10, row 1, column 2 = 15, | 9. (a) North depot = 176, (a) South depot =
180, — |
| | 10. (a) $x = 30$, $y = 50$, $z = 20$, — |
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What is verified

27 of 30 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-REI.C.6 – *problems 7, 10*

HSN-VM.C.6–C.12 – *problems 1, 2, 3, 4, 5, 6, 8, 9*

Difficulty 1–5 of 5 across 30 tagged checks.

1. $M = \begin{bmatrix} 24 & 18 \\ 31 & 12 \end{bmatrix}$, $T = \begin{bmatrix} 19 & 22 \\ 27 & 15 \end{bmatrix}$. Find $M + T$ in bikes.

Solution: Matrices of the same shape add entry by entry; each sum is the two-day rental total for one depot and one bike type.

$$M + T = \begin{bmatrix} 24 + 19 & 18 + 22 \\ 31 + 27 & 12 + 15 \end{bmatrix} = \left[\begin{array}{|c|} \hline 43 \\ \hline 58 \\ \hline \end{array} \quad \begin{array}{|c|} \hline 40 \\ \hline 27 \\ \hline \end{array} \right] \text{ bikes}$$

2. $P = \begin{bmatrix} 6 & 10 \\ 5 & 9 \end{bmatrix}$ dollars per rental. Find $1.25P$.

Solution: A scalar multiplies every entry, so the shape does not change and no entry interacts with any other.

$$1.25P = \begin{bmatrix} 1.25(6) & 1.25(10) \\ 1.25(5) & 1.25(9) \end{bmatrix} = \begin{bmatrix} \boxed{7.5} & \boxed{12.5} \\ \boxed{6.25} & \boxed{11.25} \end{bmatrix} \text{ dollars per rental}$$

A 25% rise is the scalar 1.25 because $100\% + 25\% = 125\%$.

3. $U = \begin{bmatrix} 52 & 34 \\ 47 & 29 \end{bmatrix}$ out, $V = \begin{bmatrix} 45 & 30 \\ 41 & 26 \end{bmatrix}$ returned. Find $U - V$.

Solution: Subtract entry by entry; each difference is the number of bikes of that type still out at that depot.

$$U - V = \begin{bmatrix} 52 - 45 & 34 - 30 \\ 47 - 41 & 29 - 26 \end{bmatrix} = \begin{bmatrix} \boxed{7} & \boxed{4} \\ \boxed{6} & \boxed{3} \end{bmatrix} \text{ bikes}$$

4. $W = \begin{bmatrix} 30 & 14 \\ 22 & 20 \end{bmatrix}$ bikes, $c = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$ dollars per rental. Find Wc .

Solution: A 2×2 times a 2×1 gives a 2×1 . Each entry pairs one row of W with the single column of c .

$$\text{North} = 30(6) + 14(10) = 180 + 140$$

$$\text{South} = 22(6) + 20(10) = 132 + 200$$

$$Wc = \begin{bmatrix} \boxed{320} \\ \boxed{332} \end{bmatrix} \text{ dollars}$$

5. $A = \begin{bmatrix} 4 & 7 \\ 2 & 5 \end{bmatrix}$ added, $B = \begin{bmatrix} 1 & 3 \\ 6 & 2 \end{bmatrix}$ retired. Find $3A - 2B$.

Solution: Scale first, then subtract entry by entry.

$$3A = \begin{bmatrix} 12 & 21 \\ 6 & 15 \end{bmatrix}, \quad 2B = \begin{bmatrix} 2 & 6 \\ 12 & 4 \end{bmatrix}$$

$$3A - 2B = \begin{bmatrix} \boxed{10} & \boxed{15} \\ \boxed{-6} & \boxed{11} \end{bmatrix} \text{ bikes}$$

The negative entry is real information: the South depot loses 6 standard bikes over the five months.

6. $D = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}$, $E = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$. Find DE in parts.

Solution: Row of D against column of E , one product per entry.

$$(1, 1) = 5(2) + 2(3) = 10 + 6$$

$$(1, 2) = 5(1) + 2(5) = 5 + 10$$

$$(2, 1) = 3(2) + 4(3) = 6 + 12$$

$$(2, 2) = 3(1) + 4(5) = 3 + 20$$

$$DE = \left[\begin{array}{|c|} \hline 16 \\ \hline 18 \\ \hline \end{array} \quad \begin{array}{|c|} \hline 15 \\ \hline 23 \\ \hline \end{array} \right] \text{ parts}$$

The inner dimensions match ($2 = 2$), and the answer inherits the outer ones: 2×2 .

7. Two standard and three electric passes cost 22 dollars; four standard and one electric cost 24 dollars. Write a matrix equation and solve.

Solution: The conditions are $2x + 3y = 22$ and $4x + y = 24$, which is

$$\begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 22 \\ 24 \end{bmatrix}$$

Solving by elimination: multiply the second equation by 3 to get $12x + 3y = 72$, then subtract the first.

$$\begin{aligned} 12x + 3y - (2x + 3y) &= 72 - 22 \\ 10x &= 50 \end{aligned}$$

So $x = 5$, and $4(5) + y = 24$ gives $y = 4$. A standard pass costs dollars and an electric pass costs dollars.

$x = 5$
$y = 4$

8. $F = \begin{bmatrix} 8 & 3 \\ 6 & 5 \end{bmatrix}$, $G = \begin{bmatrix} 7 & 4 \\ 12 & 9 \end{bmatrix}$. (a) Entry $(1, 2)$ of FG . (b) What it measures.

Solution (a): Row 1 of F against column 2 of G :

$$8(4) + 3(9) = 32 + 27$$

so the entry is

59

Solution (b) — instructor-judged. Row 1 is Saturday and column 2 is staff minutes, so the entry is the total staff time in minutes that Saturday's rentals require: 8 standard rentals at 4 minutes each plus 3 electric rentals at 9 minutes each. Full credit names the row, names the column, and says the entry totals across the two bike types. "It is 59 minutes" alone does not show the student can read a product matrix.

9. $H = \begin{bmatrix} 10 & 6 & 4 \\ 8 & 9 & 3 \end{bmatrix}$, $k = \begin{bmatrix} 6 \\ 10 \\ 14 \end{bmatrix}$. (a) Find Hk . (b) Why kH does not exist.

Solution (a): H is 2×3 and k is 3×1 ; the inner dimensions agree, so the product is 2×1 .

$$\begin{aligned} \text{North} &= 10(6) + 6(10) + 4(14) = 60 + 60 + 56 \\ \text{South} &= 8(6) + 9(10) + 3(14) = 48 + 90 + 42 \end{aligned}$$

$$Hk = \left[\begin{array}{|c|} \hline 176 \\ \hline 180 \\ \hline \end{array} \right] \text{ dollars}$$

Solution (b) — instructor-judged. In the order kH the shapes are 3×1 and 2×3 : the inner dimensions are 1 and 2, which do not agree, so no row of k can be paired with a column of H and the product is undefined. Full credit states the inner-dimension rule and applies it to these shapes. “Matrices do not commute” is not enough here — the issue is not a different answer but no answer at all.

10. 100 bikes in three classes; 20 more electric than standard; a full round of rentals at 6, 10 and 14 dollars takes 960 dollars. (a) Solve. (b) Interpret the matrix equation.

Solution (a): The three conditions are

$$x + y + z = 100, \quad -x + y = 20, \quad 6x + 10y + 14z = 960$$

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 0 \\ 6 & 10 & 14 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 100 \\ 20 \\ 960 \end{bmatrix}$$

From the second row $y = x + 20$. Substituting into the first gives $z = 80 - 2x$. Substituting both into the third:

$$\begin{aligned} 6x + 10(x + 20) + 14(80 - 2x) &= 960 \\ 6x + 10x + 200 + 1120 - 28x &= 960 \\ -12x &= -360 \end{aligned}$$

So $x = 30$, then $y = 50$ and $z = 20$:

standard,

electric,

cargo bikes.

$x = 30$

$y = 50$

$z = 20$

Solution (b) — instructor-judged. The coefficient matrix holds one row per condition (total count, the electric-versus-standard difference, total revenue) and one column per bike class. The variable column holds the three unknown counts. The constant column holds the three totals the conditions demand: 100 bikes, a difference of 20 bikes, and 960 dollars. Full credit ties rows to conditions and columns to unknowns; restating the three equations in words is not an interpretation.
